

These are just a few examples of remanufacturing industries. **Remanufacturing is not the same as recycling or repairing.** According to the remanufacturing industries council, remanufacturing is a comprehensive and rigorous industrial process by which previously sold, worn or non-functional products or components are returned like new or in better-than-new condition, and a warranty card is given with regards to performance level and quality.

What is special about remanufacturing

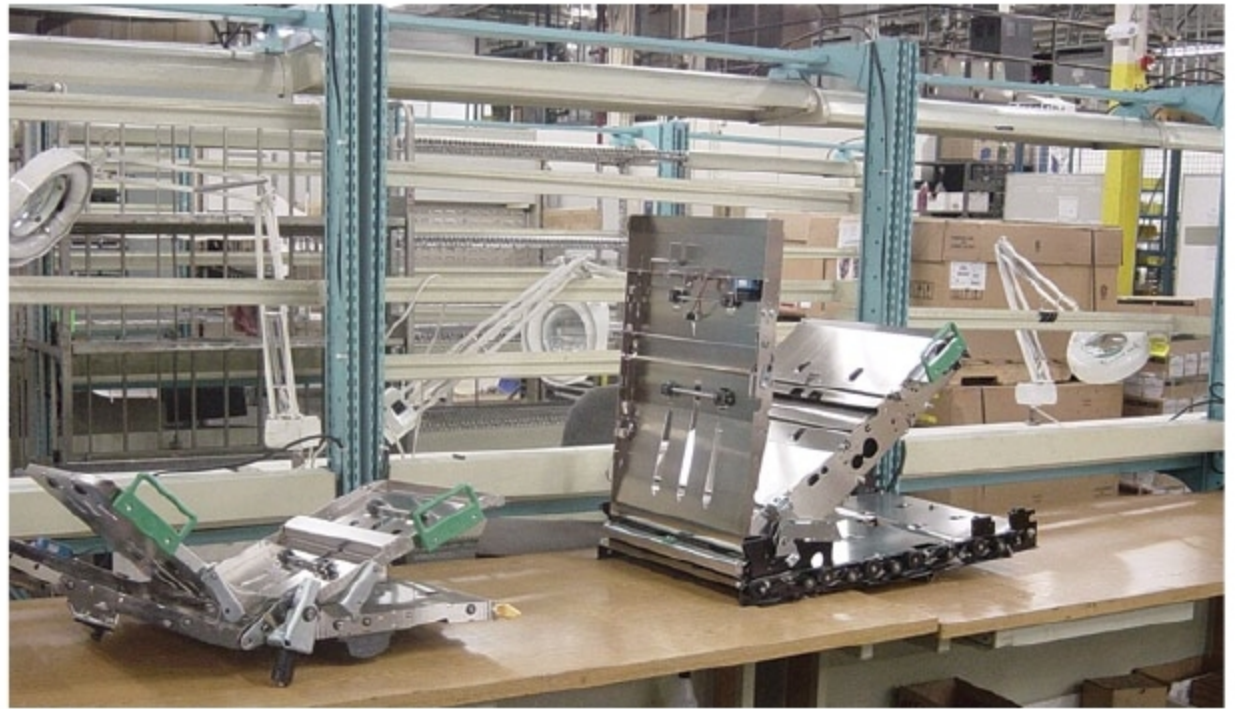
Today, the world is facing such problems as limited natural resources, sustainability, industrial pollution, environmental issues and so on. As a result, remanufacturing is becoming necessary in the industrial scenario and is slowly gaining importance in the US, China and Europe.

Manufacturers are now interested in learning how to reuse materials. One increasingly popular option is remanufacturing, which is the process of dis-assembling, cleaning and repairing components so that products can be reassembled for sale. Remanufacturing technology is predominantly used in aerospace and automotive sectors. Remanufacturing is a thorough process that returns the engine to its original specification, whereas a reconditioned engine may still contain some damaged components.

The remanufacturing sector has a European Union (EU) market potential of ninety billion euros by 2030, cementing its important role in future manufacturing. Centre for Remanufacturing and Reuse (CRR) has been established to promote the technique across Europe.

Remanufacturing is a sizable industry. For example, in the US, there are more than 73,000 companies engaged in remanufacturing. These companies employ 350,000 people and have a turnover totalling 53 billion dollars.

A report by Global Industry Analysts (GIA) says that the global remanufacturing market reached US\$ 104.8 billion by 2015. In Singapore, such sectors as aerospace, oil and gas, automotive and machinery have undertaken a number of remanufac-



Remanufacturing saves Xerox time and money (Credit: www.pierceindustries.com)

turing activities, with aerospace maintenance, repair and overhaul being the biggest contributor.

Advanced Remanufacturing and Technology Centre (ARTC), Asia's first centre for testing and developing remanufacturing technologies, was set up in Singapore in 2012. It has a public-private collaboration and aims to fill technological gaps to develop advanced remanufacturing processes. "Through this unique collaborative model, like-minded companies come together at ARTC to provide real industry requirements, spark new ideas, catalyst new opportunities and accelerate deployment of remanufacturing capabilities in a faster, better and cheaper way, than doing it alone," says David Low, chief executive officer, ARTC.

A remanufactured product consumes the lowest manufacturing time and, hence, lowers manufacturing costs. Further, product manufacturing by remanufacturing is easy to dis-assemble and rebuild. Investment in entirely new equipment is expensive and often unnecessary. Manufacturers can greatly benefit from remanufactured parts and components.

Remanufacturing helps achieve the following:

- Sustainable manufacturing
- Promotes conservation and more comprehensive utilisation of resources
- Drives the business and society to gain a circular economy

The world industry is currently shifting from a linear model to a cir-

cular one. The circular model helps recover products at the end of their lifecycle. Remanufacturing is one of the main cornerstones of this emerging circular manufacturing model. Indeed, value of remanufacturing in the UK is already 2.4 billion pounds, and it will increase to 5.6 billion pounds in the near future.

Remanufacturing spans a number of industries. It has higher chances of success in sectors where:

- Products are durable and usually contain high-value materials.
- Technology cycle is stable and longer than useful life cycle.
- Restoration technologies are available.
- Products have the potential to be leased and delivered as service rather than as hardware.

Automation is the future of the remanufacturing industry

Due to the arrival of Industry 4.0, the Internet of Things (IoT), cyber-physical systems, cloud manufacturing and so on, remanufacturing is undergoing significant transformation, and becoming more intelligent and automated.

If products are better designed to accomplish the goals of the remanufacturing process, massive improvements and efficiencies can be obtained. Adoption of automated information and communication technologies (ICTs) in remanufacturing systems today is helping create more sophisticated cyber-physical production systems (CPPSSs).